



Open Data, Design, & Development at the Office of Natural Resources Revenue

UX intern mission accomplished: plan and participate in a design studio

February 17, 2021

The design challenge

In previous research and testing done by the Office of Natural Resources Revenue (ONRR), they'd uncovered that users were confused by the trends section on the current landing page of <https://revenuedata.doi.gov/>.

Additionally, it seemed to make sense that the Production tab also show relevant trends as was the case for the two other tabs—Revenue and Disbursements.

Your mission, should you choose to accept it...

...is to create a plan to solve the trends design challenge. As an intrepid intern, I chose to accept.

With plenty of guidance from our User Experience Design mentor, Shannon, we wrote up a plan based on Basecamp's product development framework, written by Ryan Singer—[Shape Up: Stop Running in Circles and Ship Work that Matters](#).

Shannon provided previous research on the problem and a skeleton of a slide deck used from an earlier design studio.

With all of that in tow, I began to wrap my head around what needed to be done, taking into account the information my intern colleagues should know about the challenge along with Basecamp's framework.

The design studio was spread out over three consecutive weeks, for a total of three one-hour sessions which occurred every Thursday.

The brevity of the group sessions—only 60 minutes long—meant that they needed to be well thought out to ensure the time was used efficiently and productively. In other words, a good plan was necessary.

So, here's what we came up with:

- Session 1: Background Info Review (understanding the problem, determine appetite).
- Offline work after Session 1: Sketching.
- Session 2: Sketch Review & Hole Poking.
- Offline work after Session 2: hole poking.
- Session 3: Review Hole Poking

Takeaways from planning

Here are some of my learnings from the planning process using Shape Up that I found particularly interesting as a UX apprentice:

The Fat Marker Sketch. The Fat Marker sketch is a way to quickly concept or ideate a solution without getting unnecessarily caught in the details at too early of a stage in designing.

Appetite. “We use the appetite as a creative constraint on the design process,” says Singer. Design can go on and on and on; we need constraints and a scope that fits the challenge at hand. For our challenge, we’d determined it a proper scope would be three one-week sprints worth of effort.

Hole-poking. Our hole-poking was based off of “Chapter 5: Risks and Rabbit Holes” from Shape Up. As a group, we looked for rabbit holes and de-risked our solution by coming up with patches for those risks.

The Process

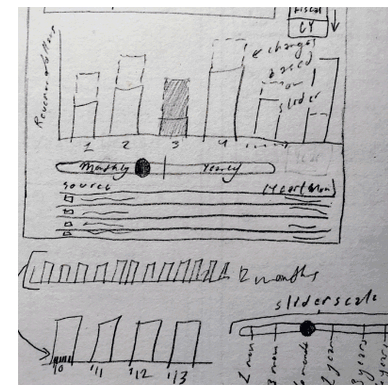
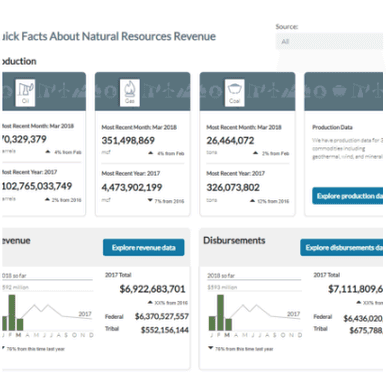
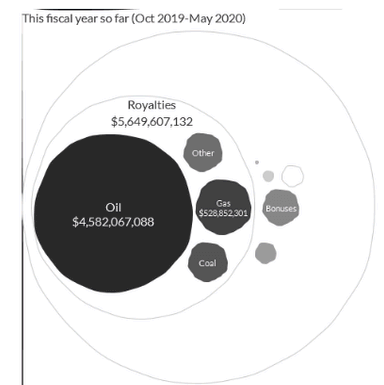
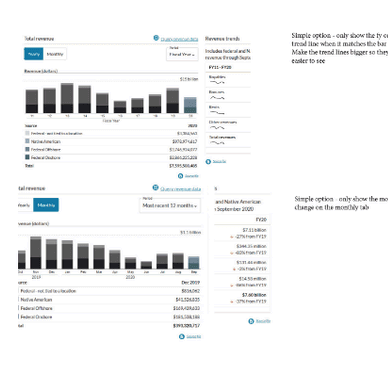
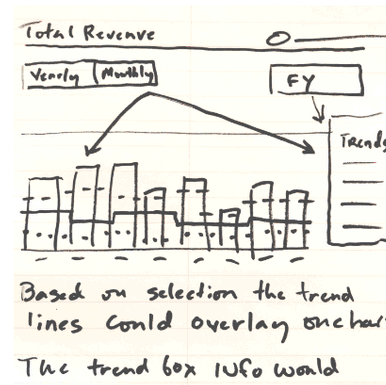
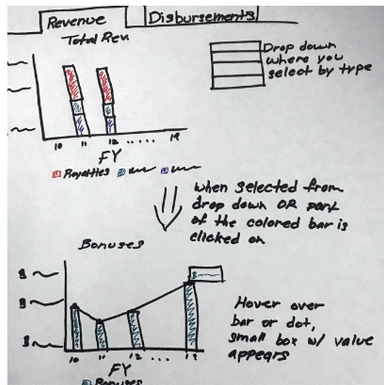
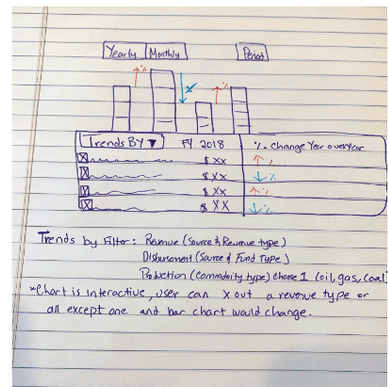
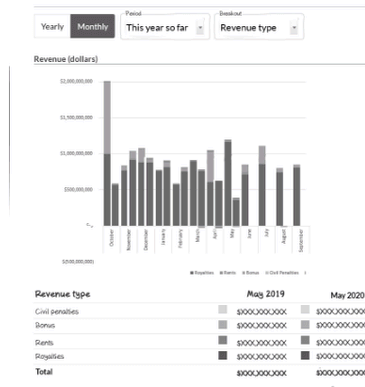
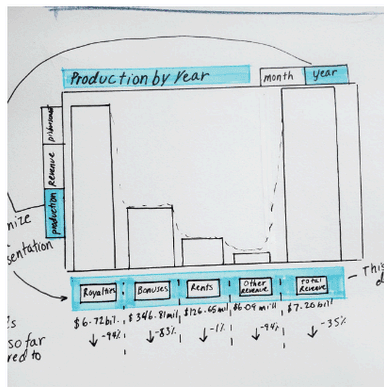
Problem Statement

In Session 1, after reviewing background information and research, we came to agreement on the problem statement:

“Domain Learners and Storytellers want to see trends over time. We know that the current trends on the homepage are confusing because they combine both monthly and yearly data. We also don’t have trends available for production.”

Sketching solutions

After Session 1, we were given the assignment to sketch out solutions.

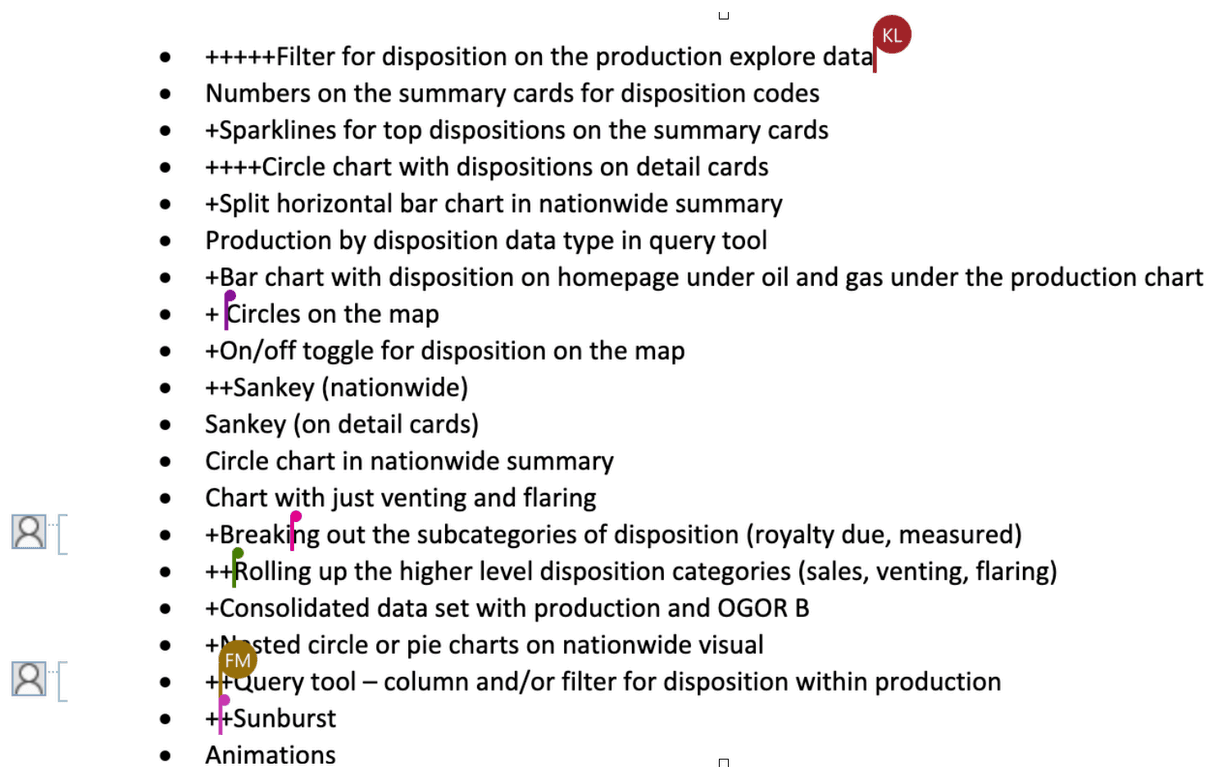


This part of the process was done independently which I thought was smart. Though collaboration is important, time alone to think through ideas gives us the room and space we need to provide meaningful solutions. We reviewed each of the solutions in Session 2.

Feature voting

Interestingly, with all of the tools available out there that help design and development teams conduct surveys, polls, etc. ONRR elected to use a simple

platform: a document which listed out the individual features of our solutions.

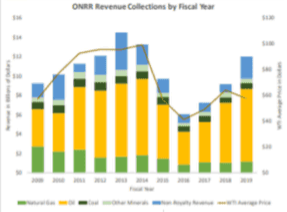
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- The screenshot shows a list of features with several colored sticky notes (KL, FM, and two unlabeled person icons) placed over it. The list is as follows:
- +++++Filter for disposition on the production explore data
 - Numbers on the summary cards for disposition codes
 - +Sparklines for top dispositions on the summary cards
 - ++++Circle chart with dispositions on detail cards
 - +Split horizontal bar chart in nationwide summary
 - Production by disposition data type in query tool
 - +Bar chart with disposition on homepage under oil and gas under the production chart
 - + Circles on the map
 - +On/off toggle for disposition on the map
 - ++Sankey (nationwide)
 - Sankey (on detail cards)
 - Circle chart in nationwide summary
 - Chart with just venting and flaring
 - +Breaking out the subcategories of disposition (royalty due, measured)
 - ++Rolling up the higher level disposition categories (sales, venting, flaring)
 - +Consolidated data set with production and OGOR B
 - +Nested circle or pie charts on nationwide visual
 - ++Query tool – column and/or filter for disposition within production
 - ++Sunburst
 - Animations

After creating this list, all of us synchronously jumped on the document to put a plus symbol next to the features we thought were important for the solution. The features with the most plus symbols were kept.

It was a simple and democratic process.

Hole-poking

For hole-poking, we were given a matrix—a simple spreadsheet—which we added our thoughts into. This was the independent work we were assigned to in Session 2. We then convened later in Session 3 as a group to review the matrix.

	A	C	D	E	F	G	H
	Feature	Is it possible within 2 sprints?	Did we miss anything?	Are we making technical assumptions that aren't fair?	Does this require new technical work we've never done before?	Are we making assumptions about how the parts fit together?	Is there a hard we should sett advance so it d trip up the design/build?
1	++++Trend line on the chart						
2							
3	++++Drop down with land type and revenue type and commodity		MF: do we need commodity option for revenue ? Which commodities would we include? The monthly fact sheet does not include commodities for revenue. SM: Nathan's bar chart has a commodity breakout.				
4	++++Land type/source drop-down for disbursements						
	++++Hovering with the value	JL: Yes	MF: the table would change as we hover				BN: Where will At the top? the

On the Y-axis were the most popular features selected from the previous stage of voting. On the X-axis were the hole-poking questions which referred to each of those specific features. For example, for the feature of “Removing Categories from the chart” we could ask the hole-poking question, “What else do we need to learn?”

Design mission outcome

With the sum of individual and cooperative efforts from the design studios, Shannon was able to create an initial wireframe. It was then iterated on from the team’s feedback.

Revenue

The amount of money collected from energy and minerals on property owned by the federal government and Native Americans.

Total revenue [Query revenue data](#) [Revenue by location](#)

Yearly

Monthly

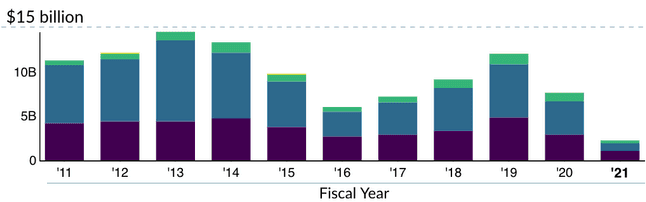
Period

Fiscal Year

Breakout

Source

Revenue by fiscal year (dollars)



Comparison

Compares data for the selected fiscal year to the previous fiscal year.

Source	FY 2021 so far (Oct - Jan)	FY 2020 so far (Oct - Jan)	Change
Federal - not tied to a location	\$633,602	\$576,010	↑ 10%
Native American	\$273,733,040	\$463,612,004	↓ (41)%
Federal Offshore	\$828,936,496	\$1,808,396,151	↓ (54)%
Federal Onshore	\$1,125,291,620	\$1,184,041,618	↓ (5)%
Total	\$2,228,594,758	\$3,456,625,783	↓ (36)%

[Source file](#)

Conclusion

The design studio process was satisfying in its organization, rationale, and capacity to have all our voices heard whether or not we were designers.

And remarkable was the ease in which it was done remotely and collaboratively, using simple platforms.

And, of course, we accomplished our goal: to create a prototype aimed at helping users better understand the trends section.

Our next step is to test this design with users.

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From the team that works on Natural Resources Revenue Data and ONRR.gov. Have feedback or a blog post idea? Drop us a line.

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